

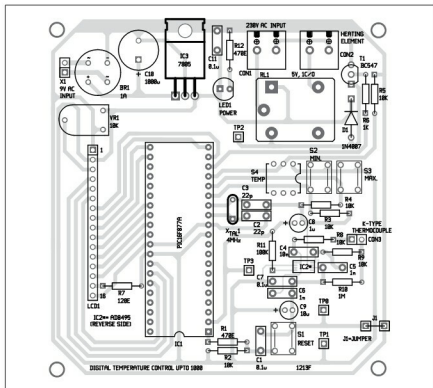


Fig. 6: Component layout for the PCB

along with MPLAB to generate hex code. The generated hex code is burnt into the microcontroller using a suitable programmer with configuration bit setting as shown in Fig. 4. The program is well commented and easy to understand.

Construction and testing

An actual-size, single-side PCB for the digital temperature controller is shown in Fig. 5 and its component layout in Fig. 6. Assemble the circuit on a PCB to save time and minimise assembly errors. Carefully assemble the components and double-check for any overlooked error. Use proper IC base for IC1. IC2 being an SMD chip, it needs to be soldered at the solder side of PCB.

After assembling and wiring the circuit properly, connect 230V, 50Hz mains supply to primary winding of the transformer, and connect transformer secondary to the PCB at X1.

Set any minimum and maximum temperature with the help of LCD display, by pressing switches S2 and S3. The maximum temperature will start from minimum temperature +10 degrees. If the temperature being measured is more than the preset maximum temperature, relay RL1 de-energises and the heating element is switched off. Similarly, when the measured temperature goes below the preset minimum temperature, relay RL1 is energised and the heating element is turned on.

PARTS LIST

Semiconductors:

- IC1 - PIC16F877A microcontroller
- IC2 - AD8495 thermocouple amplifier
- IC3 - 7805, 5V regulator
- T1 - BC547 npn transistor
- D1 - 1N4007 rectifier diode
- LCD1 - 16x2 line LCD display
- BR1 - 1A bridge-rectifier module
- LED1 - 5mm LED

Resistors (all 1/4-watt, 5% carbon):

- R1, R12 - 470-ohm
- R2-R5, R8, R9 - 10-kilo-ohm
- R6 - 1-kilo-ohm
- R7 - 120-ohm
- R10 - 1-mega-ohm
- R11 - 100-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1, C7, C11 - 0.1µF ceramic disk
- C2, C3 - 22pF ceramic disk
- C4 - 10nF ceramic disk
- C5, C6 - 1nF ceramic disk
- C8 - 1µF, 16V electrolytic
- C9 - 10µF, 16V electrolytic
- C10 - 1000µF, 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- S1-S3 - Tactile switch
- S4 - On/off switch
- RL1 - 5V, 1C/O relay
- X_{1,AL} - 4MHz crystal oscillator
- CON1, CON2 - 2-pin, 5mm terminal connector
- CON3 - 2-pin connector

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

To check the circuit for proper functioning, verify 5V power supply at TP1 with respect to TP0. TP2 is 'low' when temperature goes below minimum temperature and remains in that state until the maximum temperature is reached. ●

The author is a B.Tech (electronics engineering) from Dr A.I.T., Bengaluru



For More
Test and Equipment Stories
Visit www.electronicstest.com